

User Study Evaluation Report

Methodology, instruments, and findings template — [DATA REQUIRED]

CSE 347/652 | DES 306/525 Usable Security and Privacy
IIIT-Delhi · Monsoon 2026 · Instructor: Dr. Arun Balaji Buduru

Phase 1: User Studies to evolve Requirements, Design and Workflows
Submission: 4 October 2026

Team: [INSERT TEAM MEMBER NAMES AND ROLL NUMBERS]

Primary research status: instruments complete; fieldwork data not yet collected. All placeholders are marked [DATA REQUIRED]. No primary data has been fabricated.

User Study Evaluation Report (Phase 1)

Status: instruments complete; fieldwork results **[DATA REQUIRED]**.

This document is the standalone deliverable required by the course brief ("User Study Evaluation reports"). It consists of report Sections 9–11 (methodology, primary research, findings template) and Appendices A–E. It intentionally contains no fabricated participants, counts, percentages, averages, quotes or observations. When the studies have been run, replace every placeholder using the analysis framework in Section 9.6 and the sheets in `Data_Collection_Template.xlsx`, then regenerate this PDF.

Contents:

1. Research methodology, questions, sampling, ethics, procedure, analysis framework (Section 9)
2. Primary research instruments summary and participant information table (Section 10)
3. Findings template: quantitative, Likert, ranking, observation timings, qualitative themes, affinity, pain points vs hypotheses H1–H9, security/privacy/usability/accessibility concerns, design opportunities (Section 11)
4. Appendix A — Student questionnaire
5. Appendix B — Security guard interview protocol
6. Appendix C — Consent statement
7. Appendix D — Contextual inquiry protocol
8. Appendix E — Data collection template (XLSX, separate file)

9. User Research Methodology

9.1 Approach

The research follows a mixed-methods design-research approach as required by the course brief: quantitative data to size problems (how often, how long, how many) and qualitative data to understand why they occur and how people work around them. Three primary methods are triangulated with secondary research so that no single source drives a requirement.

Method	Population	Data type	Primary purpose
Online questionnaire (Appendix A)	Students	Quantitative (Likert, frequency, ranking) + short qualitative	Size the problem; measure attitudes to security, privacy, technology options, accessibility
Semi-structured interviews (Appendix B)	Security guards; security office/admin if accessible	Qualitative	Understand real verification workflow, exceptions, workarounds, information needs, constraints
Contextual inquiry / non-participant observation (Appendix D)	Gate environment	Quantitative timing + qualitative field notes	Ground truth for step counts, timing, queueing, environment, deviations from stated procedure
Secondary research (Section 12)	Literature, standards, existing systems	Evidence from publications	Threats, technology properties, usable-security and privacy principles

9.2 Research questions

RQ	Question	Answered by
RQ1	How does campus entry currently work in practice, and where does it deviate from the intended procedure?	Interviews, observation
RQ2	How frequently do students forget, lose, or have trouble with their ID, and what happens when they do?	Questionnaire, interviews
RQ3	How long does entry take, and when do queues form?	Observation, questionnaire
RQ4	What security weaknesses do guards and students perceive, and which have they witnessed?	Questionnaire, interviews
RQ5	What information do guards need to make a confident decision, and what information should not be shown to them?	Interviews, observation
RQ6	What are students' privacy concerns about an electronic entry system, and what would make them trust it?	Questionnaire
RQ7	Which authentication methods (card tap, phone tap, phone QR, other) do students prefer and why? What proportion has a compatible phone?	Questionnaire
RQ8	What accessibility needs exist at the gate?	Questionnaire, observation
RQ9	How are visitors handled, and what data is collected about them?	Interviews, observation
RQ10	What happens when the guard, the device, or the network is unavailable?	Interviews, observation

9.3 Sampling plan

- Students: convenience sample recruited through class groups, hostel groups and notice boards. Target: at least 40 responses across programmes and years; minimum viable: 25. Actual: [INSERT NUMBER OF PARTICIPANTS].
- Security guards: purposive sample. Target: 3–6 guards across shifts (morning peak, afternoon, night) and gates. Actual: [INSERT NUMBER OF PARTICIPANTS]. Permission from the security supervisor must be obtained before approaching guards on duty; interviews should be scheduled off-shift or during quiet periods.
- Administration / security office: 1–2 interviews if access is granted. Actual: [INSERT NUMBER / "not accessible"].
- Observation: at least 3 sessions of 30–45 minutes each, covering one morning peak, one off-peak, and one evening/hostel-return period, ideally at two different gates. Actual: [INSERT SESSIONS CONDUCTED].

9.4 Ethics

- Participation is voluntary and anonymous; no names, roll numbers, phone numbers or e-mail addresses are collected in the questionnaire.
- Interviews are recorded only in written notes unless the participant explicitly consents to audio; audio, if any, is deleted after transcription.
- Observation is non-participant and non-interfering: observers do not speak to people in the queue, do not record faces, names, or card details, and stop if asked by security staff.

- Guards are a vulnerable population in the employment sense (contract staff, may fear criticism). The consent statement makes clear that findings are reported in aggregate, that no individual guard's performance is assessed, and that participation cannot affect their employment.
- The consent statement (Appendix C) is read aloud before interviews and shown as the first page of the questionnaire.
- Data is stored on a password-protected team drive accessible only to team members and deleted at the end of the course.

9.5 Data collection procedure

Step	Activity	Output	Responsible
1	Obtain permission from security supervisor / relevant office for interviews and observation	Written or e-mail permission	Team lead
2	Pilot questionnaire with 3–5 students; fix ambiguous wording	Revised form	Team
3	Distribute questionnaire (Google Forms); leave open 7 days	Response CSV	Team
4	Conduct observation sessions using Appendix D checklist	Field notes + timing sheets in Data_Collection_Template.xlsx	Two observers per session
5	Conduct guard interviews using Appendix B protocol	Notes template per interview	Interviewer + note-taker
6	Conduct admin interview if available	Notes	Team lead
7	Transcribe / clean; enter into Data_Collection_Template.xlsx	Cleaned dataset	Team
8	Quantitative analysis: frequencies, Likert medians and distributions, ranking scores	Tables and charts	Team
9	Qualitative analysis: open coding of interview and open-text responses; affinity clustering into themes	Affinity map, theme table	Whole team, 1 session
10	Map findings to user needs and requirements; update traceability matrix	Updated Section 37 and Appendix F	Team

9.6 Analysis framework

Quantitative

- Frequency distributions for every categorical item (bar charts).
- For Likert items (5-point): report median, mode, and full distribution; do not report means as the sole summary. Group "agree" and "strongly agree" only when the full distribution is also shown.
- For ranking items: Borda-count style scores (rank 1 = n points) plus first-choice frequency.
- Cross-tabulations of interest: authentication-method preference vs. smartphone ownership; privacy concern vs. trust; year of study vs. forgotten-ID frequency.
- Observation: mean, median and range of per-person processing time, split by "uncontested" vs "questioned"; queue length at fixed intervals.

Qualitative

- Open coding of all interview notes and open-text questionnaire answers by two team members independently; codes reconciled.
- Affinity diagramming: codes clustered into themes; each theme labelled with a user-voice statement.
- Themes classified into: pain point, security concern, privacy concern, usability concern, accessibility concern, design opportunity.
- Each theme linked to at least one requirement or explicitly marked as out of scope.

Evidence labelling convention used in this report

Label	Meaning
[PRIMARY]	Finding from our own questionnaire, interviews or observation. Must cite the question/session.
[SECONDARY]	Finding from a cited published source.
[ASSUMPTION]	Team judgement not yet backed by evidence; must be validated.
[DATA REQUIRED]	Slot for primary evidence not yet collected.

10. Primary Research

10.1 Student Questionnaire

Full instrument with question types, options, required flags and purposes is in Appendix A. Summary: 20 questions in six sections (Background; Current Entry Experience; Problems and Failures; Security Perceptions; Privacy and Trust; Preferences, Accessibility and Features). Estimated completion time 6–8 minutes. Designed for Google Forms. No personally identifying information collected.

10.2 Security Guard Interviews

Full protocol with introduction, consent statement, 14 core questions, follow-up probes and notes template is in Appendix B. Semi-structured, 20–30 minutes, conducted off-shift or during a quiet period with supervisor permission. The same protocol, with the administrative block (Q13–Q14) expanded, is used for the security office / ID cell interview if access is obtained.

10.3 Contextual Inquiry

Full protocol with observation checklist, timing method, what-to-record / what-not-to-record lists and ethical constraints is in Appendix D. Non-participant observation from a position agreed with security staff. Two observers per session: one timing, one taking field notes.

10.4 Participant Information

All participant information is reported in aggregate only.

Population	Target	Actual	Notes
Students (questionnaire)	≥40	[INSERT NUMBER OF PARTICIPANTS]	Programme and year distribution: [INSERT DISTRIBUTION]
Security guards (interview)	3–6	[INSERT NUMBER OF PARTICIPANTS]	Shifts and gates covered: [INSERT]
Administration (interview)	1–2	[INSERT NUMBER / "not accessible"]	Office: [INSERT]
Observation sessions	≥3	[INSERT NUMBER]	Dates, times, gates: [INSERT]

10.5 Data Collection Procedure

As described in Section 9.5. Actual dates: questionnaire open [INSERT DATES]; interviews conducted [INSERT DATES]; observation sessions [INSERT DATES]. Deviations from plan: [INSERT or "none"].

11. Primary Research Findings

Status: [DATA REQUIRED]. This section is a structured template. No primary data has been collected at the time of writing. Every value below is a placeholder. Nothing in this section may be cited as a finding until the placeholders are replaced with actual results from Appendix E (Data_Collection_Template.xlsx). The team must also delete any placeholder row for which no data exists rather than estimate.

11.1 Quantitative Findings

11.1.1 Participant profile

- Total valid responses: [INSERT ACTUAL NUMBER]
- Programme: UG [n] / PG [n] / PhD [n]
- Year of study distribution: [INSERT DISTRIBUTION]
- Hosteller / day scholar: [n] / [n]
- Smartphone ownership (Q17): [INSERT ACTUAL PERCENTAGE] own a smartphone; [INSERT ACTUAL PERCENTAGE] report NFC capability; [INSERT ACTUAL PERCENTAGE] don't know

11.1.2 Frequency distributions

Item	Question	Result
Entries per day	Q3	[INSERT DISTRIBUTION]
How ID is carried	Q4	[INSERT DISTRIBUTION]
Guard actually looks at card	Q5	[INSERT DISTRIBUTION]
Forgotten ID in last month	Q6	[INSERT ACTUAL PERCENTAGE] at least once
What happened when ID forgotten	Q7	[INSERT DISTRIBUTION]
Ever lost ID	Q9	[INSERT ACTUAL PERCENTAGE]
Days to replacement	Q10	median [INSERT ACTUAL VALUE]
Typical wait at peak	Q11	[INSERT DISTRIBUTION]
Witnessed someone enter with another's ID / no check	Q12	[INSERT ACTUAL PERCENTAGE]
Preferred method (first choice)	Q16	[INSERT DISTRIBUTION]

11.1.3 Likert-scale summaries

Report median, mode and full 5-point distribution for each.

Statement	Q	SD	D	N	A	SA	Median
"The current ID check reliably stops unauthorised people"	Q13a	[n]	[n]	[n]	[n]	[n]	[x]
"I am concerned that a lost card could be used by someone else"	Q13b	[n]	[n]	[n]	[n]	[n]	[x]
"I would be comfortable with the college recording the time I enter campus"	Q14a	[n]	[n]	[n]	[n]	[n]	[x]
"I would want to see my own entry records"	Q14b	[n]	[n]	[n]	[n]	[n]	[x]
"I would trust an electronic system more than a visual check"	Q15a	[n]	[n]	[n]	[n]	[n]	[x]
"I am worried that entry data could be used for purposes other than security"	Q15b	[n]	[n]	[n]	[n]	[n]	[x]
"Retrieving my ID card at the gate is inconvenient"	Q11b	[n]	[n]	[n]	[n]	[n]	[x]

11.1.4 Ranking results (Q16: preferred authentication method)

Option	Rank-1 count	Borda score
Tap physical smart card	[n]	[x]
Tap phone (NFC)	[n]	[x]
Show QR code on phone	[n]	[x]
Keep current visual check	[n]	[x]
Other (specify)	[n]	[x]

11.1.5 Observation timings (Appendix D)

Measure	Peak session	Off-peak session	Evening session
People observed	[n]	[n]	[n]
Median time from reaching guard to passing (uncontested)	[x s]	[x s]	[x s]
Median time when questioned	[x s]	[x s]	[x s]
Proportion questioned	[x %]	[x %]	[x %]
Proportion waved through without visible card	[x %]	[x %]	[x %]
Max queue length observed	[n]	[n]	[n]
Visitor register entries observed and median time	[n], [x s]	[n], [x s]	[n], [x s]

11.2 Qualitative Findings

Interview and open-text responses coded and clustered. Each theme must be supported by at least one real, anonymised quote with participant code (e.g., G2 = guard 2; S14 = student response 14).

Theme code	Theme (user-voice label)	Sources	Representative quote
T1	[INSERT THEME]	[G1, G3, S7...]	"[INSERT REAL PARTICIPANT QUOTE]"
T2	[INSERT THEME]		"[INSERT REAL PARTICIPANT QUOTE]"
T3	[INSERT THEME]		"[INSERT REAL PARTICIPANT QUOTE]"
...			

11.3 Affinity Themes

Photograph or export of the affinity map: [INSERT IMAGE / "see Appendix E"]. Top-level clusters and the number of codes in each: [INSERT].

11.4 Pain Points

Each hypothesis from Section 8 is confirmed, refuted, or modified here.

Hypothesis	Verdict	Evidence
H1 Card retrieval delays queue	[CONFIRMED / REFUTED / MODIFIED]	[INSERT OBSERVED FINDING, e.g., median retrieval time; Q11]
H2 Expiry date not read	[...]	[INSERT]
H3 Facial familiarity replaces card check	[...]	[INSERT]
H4 Forgotten ID common, inconsistent handling	[...]	[INSERT]
H5 Lost cards not blocked; slow replacement	[...]	[INSERT]
H6 Visitor register slow and leaks phone numbers	[...]	[INSERT]
H7 Peak pressure leads to waving through	[...]	[INSERT]
H8 No record for incident investigation	[...]	[INSERT]
H9 Students uneasy about data use	[...]	[INSERT]
New pain points not hypothesised		[INSERT]

11.5 Security Concerns

Concerns raised by participants (not by the team). [DATA REQUIRED]

Concern	Raised by	Frequency / strength	Links to requirement
[INSERT]	Students / Guards / Admin	[INSERT]	SEC-xx

11.6 Privacy Concerns

[DATA REQUIRED]

Concern	Raised by	Frequency / strength	Links to requirement
[INSERT]			PRIV-xx

11.7 Usability Concerns

[DATA REQUIRED]

Concern	Raised by	Frequency / strength	Links to requirement
[INSERT]			USE-xx

11.8 Accessibility Concerns

[DATA REQUIRED] (Q18, observation)

Need	Raised by	Links to requirement
[INSERT]		USE-xx / NFR-xx

11.9 Design Opportunities

Derived from confirmed pain points and themes. [DATA REQUIRED]

Opportunity	Supporting finding	Addressed in
[INSERT]	T-x / H-x	Section / Screen

11.10 How findings will be used

Once fieldwork data replaces the placeholders above, the team will (1) update the "User evidence" column of every requirement in Sections 19–23, (2) update the Design Decision Matrix (Section 34), (3) revise the personas (Section 16) to reflect actual distributions, and (4) re-run the traceability check in Section 37 so that every requirement cites either a primary finding, a secondary source, or a labelled assumption. Until then, all requirements are grounded in secondary research and labelled assumptions.

Appendix A. Student Questionnaire

Title for Google Forms: Campus Entry Experience Survey (Usable Security & Privacy course project)

Form description (shown on page 1):

This short survey (6–8 minutes) is part of a student course project in CSE 347/652 | DES 306/525 Usable Security and Privacy at IIIT-Delhi. We are studying how campus entry works today and how it could be made more secure, private and convenient. The survey is anonymous: we do not ask for your name, roll number, e-mail or phone number. Responses are stored on a password-protected team drive, reported only in aggregate, and deleted at the end of the course. Participation is voluntary and you may stop at any time. By clicking "Next" you confirm you are 18 or older and agree to take part. Questions: contact the team at [TEAM CONTACT E-MAIL].

Google Forms settings: Do not collect e-mail addresses. Do not limit to one response (requires sign-in). Shuffle option order OFF (ranking and Likert need fixed order). Show progress bar ON.

Section 1 — Background

#	Question	Type	Options	Required	Purpose
Q1	Which programme are you in?	Multiple choice	UG (B.Tech) / PG (M.Tech / MSc) / PhD / Other	Yes	Segment respondents; check sample balance
Q2	Are you a hosteller or a day scholar?	Multiple choice	Hosteller / Day scholar / Other	Yes	Entry frequency and time-of-day patterns differ
Q3	On a typical weekday, how many times do you pass through a campus entry gate?	Multiple choice	0–1 / 2–3 / 4–5 / More than 5	Yes	Size how often authentication happens per user; informs speed requirement

Section 2 — Current Entry Experience

#	Question	Type	Options	Required	Purpose
Q4	How do you usually carry your college ID card?	Multiple choice	On a lanyard, visible / In wallet or pocket / In bag / I often don't carry it / Other	Yes	Retrieval effort; informs H1 and form-factor decision
Q5	When you enter, how closely does the guard usually look at your ID?	Multiple choice	Does not look; I walk through / Glances at the card from a distance / Looks at the card and my face / Asks me to stop and checks carefully / Varies a lot	Yes	Measures perceived rigour of current check (RQ1, H3, H7)
Q11a	At the busiest time you enter, how long do you typically wait in the queue at the gate?	Multiple choice	No wait / Under 30 seconds / 30 s – 1 min / 1–3 min / More than 3 min	Yes	Quantifies queueing (RQ3)
Q11b	"Retrieving my ID card at the gate is inconvenient."	Likert (5)	Strongly disagree / Disagree / Neutral / Agree / Strongly agree	Yes	Perceived friction of current mechanism

Section 3 — Problems and Failures

#	Question	Type	Options	Required	Purpose
Q6	In the last month, how many times did you arrive at the gate without your ID card?	Multiple choice	0 / 1 / 2–3 / 4 or more	Yes	Frequency of forgotten-ID case (H4)
Q7	The last time you arrived without your ID, what happened?	Checkboxes (multiple)	Guard let me in without asking / Guard recognised me and let me in / I showed another ID or proof / Guard asked questions then let me in / Someone vouched for me / I was refused entry / Never happened to me / Other (specify)	Yes	Documents actual fallback behaviour (RQ2)
Q8	If you were refused entry without your ID, what did you do?	Short answer	—	No	Qualitative detail on the refusal path
Q9	Have you ever lost your college ID card?	Multiple choice	Yes, once / Yes, more than once / No	Yes	Frequency of lost-credential case (H5)
Q10	If yes, roughly how many days did it take to get a replacement, and could you enter campus in the meantime?	Short answer	—	No	Replacement latency and interim access (H5)
Q12	Have you ever seen someone enter the campus using another person's ID, a fake ID, or without any check?	Multiple choice	Yes, more than once / Yes, once / Not sure / No	Yes	Witnessed security failures without asking respondent to self-incriminate (RQ4)

Section 4 — Security Perceptions

#	Question	Type	Options	Required	Purpose
Q13a	"The current ID check reliably stops unauthorised people from entering."	Likert (5)	SD / D / N / A / SA	Yes	Perceived effectiveness (RQ4)
Q13b	"I am concerned that a lost or stolen ID card could be used by someone else to enter."	Likert (5)	SD / D / N / A / SA	Yes	Perceived risk of credential theft
Q13c	Which of the following would you consider a security problem with the current system? (select all)	Checkboxes	Cards can be borrowed / Cards can be faked / Lost cards keep working / Expiry is not checked / Visitors are not checked properly / Guards are too busy to check / None of these / Other	Yes	Maps perceived problems to threat model rows

Section 5 — Privacy and Trust

#	Question	Type	Options	Required	Purpose
Q14a	"I would be comfortable with the college recording the time I enter campus."	Likert (5)	SD / D / N / A / SA	Yes	Acceptance of entry logging (PRIV requirements)
Q14b	"I would want to be able to see my own entry records."	Likert (5)	SD / D / N / A / SA	Yes	Transparency need (PRIV-08)
Q14c	Which of your details would you be comfortable with a gate guard seeing on a screen when you enter? (select all)	Checkboxes	Photo / First name / Full name / Roll number / Programme / Hostel / Phone number / None of these	Yes	Directly informs guard-screen data minimisation (PRIV-03, screen 5)
Q15a	"I would trust an electronic check (card tap / phone) more than the current visual check."	Likert (5)	SD / D / N / A / SA	Yes	Baseline trust in electronic authentication
Q15b	"I am worried that entry data could be used for purposes other than campus security (e.g., attendance, tracking)."	Likert (5)	SD / D / N / A / SA	Yes	Purpose-limitation concern (PRIV-02)
Q15c	What would make you trust an electronic entry system more?	Short answer	—	No	Qualitative trust factors

Section 6 — Preferences, Accessibility and Features

#	Question	Type	Options	Required	Purpose
Q16	Rank these ways of proving who you are at the gate, from most preferred (1) to least (5).	Ranking grid (Google Forms: multiple-choice grid, one column per rank)	Tap a physical smart card / Tap my phone (NFC) / Show a QR code on my phone / Keep the current visual ID check / Other	Yes	Method preference (RQ7); drives decision matrix D1
Q17	Which best describes your phone?	Multiple choice	Smartphone with NFC (tap-to-pay works) / Smartphone, but I don't know if it has NFC / Smartphone without NFC / I do not carry a smartphone regularly	Yes	Feasibility of phone-based credential (A3)
Q18	Do you have any need that affects how you use the gate (e.g., difficulty reaching into a bag, low vision, hearing, using a wheelchair, carrying a child or heavy equipment)? Optional; do not include anything you'd rather not share.	Short answer	—	No	Accessibility requirements (RQ8). Framed to avoid disclosing a diagnosis
Q19	If your ID stopped working at the gate, what would you expect to happen?	Checkboxes	Guard checks my photo and lets me in / I verify with a one-time code on my phone / I go to the security office / I am not allowed in until fixed / Other	Yes	Expectations for failure handling (FR-13, screen 12)
Q20	Is there anything else about entering campus, security, or privacy at the gate that you want to tell us?	Paragraph	—	No	Open qualitative capture

Question numbering note: Q11a/b appear in Section 2 for flow but are numbered to match the analysis tables in Section 11.

Pilot instructions: Run with 3–5 students. Check: completion time under 8 minutes; no question interpreted two ways; ranking grid works on mobile; option lists are exhaustive (watch "Other" frequency).

Appendix B. Security Guard Interview Protocol

Format: Semi-structured, 20–30 minutes, one interviewer and one note-taker. Conducted off-shift or during a quiet period, with permission from the security supervisor. Language: Hindi or English as the participant prefers; the protocol should be translated verbally where needed and the translation noted.

B.1 Introduction (read aloud)

Namaste / Hello. Thank you for speaking with us. We are students in a course on usable security at IIIT-Delhi. We are studying how campus entry works so that any future system is designed around the people who actually operate it. We want to learn from your experience; there are no right or wrong answers, and we are not evaluating you or your work. This will take about 20–30 minutes.

B.2 Consent statement (read aloud; see also Appendix C)

Your participation is voluntary. You can skip any question or stop at any time, and this will have no effect on your job. We will not record your name, employee ID, or which gate you work at in our report. We will take written notes. We will only record audio if you say yes, and we will delete the audio after we write it up. What you tell us will be combined with what other guards tell us and reported as a group, so no one can be identified. Do you agree to take part? [Record: Yes / No]
May we record audio? [Yes / No]

B.3 Core questions

#	Question	Topic	Follow-up probes
Q1	Can you walk me through what you do when a student comes to the gate, from the moment you see them to when they're through?	Current workflow	What do you look at first? How long does it usually take? Does it differ at the vehicle gate?
Q2	How do you decide the ID is genuine?	Verification	What features do you look for? Have you seen cards that looked wrong? What did you do?
Q3	How often do you recognise students by face rather than checking the card?	Verification / workaround	Is that more common at certain times? Do you feel that is reliable?
Q4	Do you check the validity or expiry date on the card?	Expiry	How easy is it to read? What happens if it's expired?
Q5	What do you do if the photo doesn't seem to match the person, or the card looks suspicious?	Suspicious / fake IDs	Who do you call? How long does it take? How often does this happen?
Q6	What happens when a student arrives without an ID card?	Forgotten ID	Is there a set procedure? Does it vary by guard or shift? What do students usually say?
Q7	If a student has lost their card, how would you know if someone else is using it?	Lost / stolen	Is there any list or notice you receive?
Q8	Tell me about the busiest times. What changes in how you check?	Busy periods / delays	How long do queues get? Do you ever have to speed up checks? What would help most at those times?
Q9	How do you handle visitors?	Visitors	What information do you write down? Who checks it? What happens with the register afterwards? What is hardest about visitors?
Q10	Have you used any electronic device or system at a gate here or elsewhere, e.g., a card reader, scanner, or tablet? What was your experience?	Device / network	What went wrong? What happened when the device or network failed? Was there a fallback?
Q11	If a screen showed you information about the person at the gate, what would you need to see to make a confident decision?	Information needed	Photo? Name? Roll number? Department? Hostel? Anything else?
Q12	Is there information about students that you think should not be shown to you, or that you'd be uncomfortable being responsible for?	Privacy	Phone numbers? Addresses? Why?
Q13	What training did you receive for gate duty, and how long did it take to feel confident?	Training	What would a new guard find hardest?
Q14	If you could change one thing about how entry works, what would it be?	Desired features	Why? What would you be worried about with an electronic system?

B.4 Additional questions for security office / ID cell (if accessible)

#	Question
A1	How is a new ID card issued? What documents and approvals are needed, and how long does it take?
A2	What happens when a student reports a lost or stolen card? Is the old card blocked anywhere?
A3	How are cards handled when a student graduates, withdraws, or is suspended?
A4	Who approves visitors, and how is the approval communicated to the gate?
A5	Are there areas or times with restricted access? How is that enforced today?
A6	If there is an incident (theft, intrusion), what records do you have to investigate who was on campus?
A7	Who would have administrative access to an electronic entry system, and what controls would you expect on that access?

B.5 Closing

Thank you. Is there anything you'd like to add or anything you'd prefer we not include? If you think of something later, here is how to reach us: [TEAM CONTACT].

B.6 Notes template (one per interview)

Interview code: G__ Date: ____ Duration: ____ min
 Interviewer: ____ Note-taker: ____
 Shift type (morning/afternoon/night): ____ Gate type (pedestrian/vehicle/both): ____
 Years of experience at this campus (approx.): ____ Audio consent: Y / N

 Q1 Workflow:
 Q2 Genuine check:
 Q3 Face recognition:
 Q4 Expiry:
 Q5 Suspicious ID:
 Q6 Forgotten ID:
 Q7 Lost card:
 Q8 Busy periods:
 Q9 Visitors:
 Q10 Devices/network:
 Q11 Info needed:
 Q12 Info NOT to show:
 Q13 Training:
 Q14 One change / worries:

 Verbatim quotes worth keeping (mark with ") :
 Observed emotions / emphasis:
 Contradictions with observation or other interviews:
 Follow-up needed:

Appendix C. Participant Information and Consent Statement

Project: Campus Entry Authentication System — Phase 1 User Study Course: CSE 347/652 | DES 306/525 Usable Security and Privacy, IIIT-Delhi, Monsoon 2026 Team contact: [TEAM CONTACT E-MAIL] Course instructor: Dr. Arun Balaji Buduru

What is this study? We are a student team studying how people enter the IIIT-Delhi campus and how the process could be made more secure, more private and easier to use. This is coursework, not an official institutional project.

What will I be asked to do? Depending on the activity: (a) complete an anonymous online questionnaire of about 6–8 minutes; or (b) take part in an interview of about 20–30 minutes; or (c) be present in a public area (the entry gate) where we observe the general process without recording individuals.

What information do you collect? For the questionnaire: only your answers; no name, roll number, e-mail or phone number. For interviews: written notes, and audio only if you separately agree; no name or employee ID. For observation: counts and timings of the process and general notes about the environment; no names, faces, photographs, card details or vehicle numbers.

How is it stored and for how long? On a password-protected drive accessible only to the team, reported only in aggregate, and deleted at the end of the course (December 2026).

Voluntary participation. You do not have to take part. You can skip any question or stop at any time without giving a reason. For staff: taking part or not taking part has no effect on your employment and no individual's performance is assessed or reported.

Risks and benefits. We do not anticipate any risk. There is no direct benefit or payment. The results may help improve campus entry design.

Contact. For questions about the study, contact the team. For concerns about how the study is conducted, contact the course instructor.

Consent (interview: read aloud and record verbally; questionnaire: shown as first page)

I have read / been told the information above. I understand my participation is voluntary and anonymous. I agree to take part. Interview only: I agree to audio recording: Yes / No

Appendix D. Contextual Inquiry (Observation) Protocol

Purpose. Establish ground truth for how entry actually happens: step count, timing, queue behaviour, guard actions, exceptions, and environmental constraints. Interviews tell us what people believe they do; observation shows what happens.

Type. Non-participant, non-interfering observation in a public area. Observers do not speak to people in the queue, do not handle cards, do not photograph or film, and do not ask guards questions during the session (questions go to the interview).

D.1 Pre-conditions

- Written or e-mail permission from the security supervisor; inform the guards on shift before starting.
- Observers wear visible college IDs and stand where the supervisor agrees, out of the flow.
- Stop immediately if asked by security staff or if an incident occurs.

D.2 Sessions

Session	Target window	Rationale
S1	Morning peak (e.g., 08:30–09:15)	Maximum load; queue formation; speed pressure
S2	Off-peak (e.g., 14:30–15:15)	Baseline behaviour without pressure
S3	Evening / hostel return (e.g., 21:30–22:15)	Low light; different guard shift; visitor sign-out
S4 (optional)	Second gate, any window	Compare gates

Two observers per session: Observer A times individuals; Observer B writes field notes and counts queue length every 2 minutes.

D.3 Timing method

- Use a stopwatch app with lap function.
- T0: person reaches the guard's position (front of queue). T1: person passes the gate line or is turned away. Record T1–T0 in seconds.
- Classify each timed person as: U (uncontested: card shown or not, no stop), Q (questioned: guard stops and speaks), N (no card visible), V (visitor), R (refused).
- Time at least 30 consecutive people per session, or the whole session if fewer.
- Record queue length (number waiting behind the person at the guard) at 2-minute intervals.

D.4 Observation checklist (Observer B)

Environment

- ☐ Lighting (adequate / dim / glare)
- ☐ Weather / shelter for guard and queue
- ☐ Noise level
- ☐ Position of guard relative to flow; is there a desk, booth, barrier, turnstile?
- ☐ Power outlet / network equipment visible?
- ☐ Space available for a reader or display; would a queue block it?

Process

- ☐ How does the guard signal "go ahead"? (nod, gesture, verbal, nothing)
- ☐ Where is the card when shown? (in hand, on lanyard, waved from distance, not shown)
- ☐ Does the guard ever look at the expiry date? Turn the card over?
- ☐ Does the guard compare photo to face? For how long?
- ☐ Does behaviour change when the queue grows? At what length?
- ☐ Are groups checked individually or as a group?
- ☐ Vehicle gate: what is checked and from what distance?

Exceptions (record each occurrence with time, type, duration, outcome)

- ☐ No card
- ☐ Guard questions holder
- ☐ Guard calls someone
- ☐ Refusal
- ☐ Visitor registration: steps, what is written, who writes, time taken, where the register sits and who can read it
- ☐ Any apparent workaround (e.g., friend passing card back, guard waving group)

Guard workload

- ☐ Concurrent duties (phone, vehicle barrier, register, radio)
- ☐ Number of guards present
- ☐ Interruptions

D.5 What to record

- Counts, durations, queue lengths, classification codes (U/Q/N/V/R).
- Environmental conditions and physical constraints.
- Descriptions of process and behaviour in general terms ("guard asked holder a question", not what was asked if it contained personal detail).
- Time-stamped exceptions and their outcomes.

D.6 What NOT to record

- Names, roll numbers, faces, photographs, video, audio.
- Card contents, vehicle registration numbers, visitor register contents.
- Identifying details of individual guards (name, badge, appearance).
- Conversations between guard and student beyond their general nature.
- Anything a guard or student asks you not to record.

D.7 Ethical considerations

- The gate is a public space and the observed behaviour is public; however, individuals have not consented, so the protocol collects only non-identifying aggregate data.
- Guards are informed in advance and may decline; observation of a guard who declines does not proceed.
- Observers must not create the impression of auditing or reporting on individual guards. The consent statement (Appendix C) is available to any guard who asks.
- If observers witness a security incident, they do not intervene and do not record identifying details; they note only "incident occurred, observation paused".

D.8 Field note sheet (per session)

Session: S_ Date: ____ Window: ____-____ Gate: ____ Observers: ____ / ____
 Guards present: ____ Weather/light: ____ Concurrent guard duties: ____
 Queue length at 2-min intervals: _____
 Timed persons (code, seconds): U 4, U 3, Q 22, N 15, ...
 Exceptions log:
 hh:mm type duration outcome note
 Process notes:
 Environment notes:
 Workarounds observed:
 Questions for interviews:

Timings are entered into `Data_Collection_Template.xlsx` (Appendix E), sheet "Observation".